

Jeremiah Sol Larocco

Senior Software Engineer | Full Stack & Cloud-Native Development

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PROFESSIONAL SUMMARY

Software engineer with 15+ years delivering scalable cloud-native applications and distributed systems. Proficient in full-stack development with TypeScript, Node.js, React, Spring Boot, and PostgreSQL on AWS. Owns quality, performance, and reliability across the development lifecycle. Experienced in automated testing, production debugging, REST API design, and cross-functional collaboration in agile environments.

WORK EXPERIENCE

Software Engineer

Dec/2018 - Jun/2021

Thermopylae Sciences + Technology

Onsite | Chantilly, Virginia

- Led full-stack development of **cloud-native geospatial web applications** serving federal and commercial clients, architecting **React frontends** with centralized state management and **Node.js/TypeScript services** backed by PostgreSQL.
- Designed and implemented **RESTful APIs** for map visualization and data ingestion pipelines, optimizing **PostgreSQL query performance** to deliver sub-second responses on datasets exceeding millions of geospatial features.
- Took end-to-end ownership of **production reliability**, conducting root cause analysis on complex outages and implementing durable fixes that eliminated recurring incident classes across deployed environments.
- Built comprehensive **automated test suites** spanning unit, integration, and end-to-end coverage, enabling confident releases within a two-week agile sprint cadence.
- Collaborated with product managers, designers, and engineering leadership to translate customer requirements into **secure, performance-optimized features** aligned with product strategy and usability goals.
- Deployed and maintained applications on **AWS infrastructure** using containerized services, integrating CI/CD pipelines and environment configuration through SSM parameter stores and S3 artifact storage.
- Drove **code quality standards** through active code reviews, mentoring engineers on testable design patterns and maintainable TypeScript codebases across shared product modules.
- Partnered with operations stakeholders to improve **system observability**, adding structured logging and metrics instrumentation that shortened mean time to detection for production anomalies.
- Identified and resolved **performance bottlenecks** in geospatial rendering and database access layers, improving dashboard load times for data-heavy workflows used by distributed field teams.

Software engineer

Jul/2013 - Nov/2018

SolidFire, Inc.

Onsite | Boulder, Colorado

- Developed **scalable management software** for all-flash storage clusters, building **Spring Boot services** and REST APIs that orchestrated provisioning, monitoring, and lifecycle operations across distributed node fleets.
- Engineered **high-performance backend components** in Java and Python handling concurrent storage requests, applying profiling and optimization techniques that improved throughput under peak cluster load.
- Owned **production debugging and incident response** for customer-facing management interfaces, performing root cause analysis on race conditions and timeout failures in multi-node environments.
- Implemented **automated integration and regression test frameworks** covering API contracts and storage provisioning workflows, raising release confidence across quarterly product cycles.
- Collaborated with SRE and QA teams to harden **deployment pipelines**, containerize services, and validate reliability targets before GA releases to enterprise customers worldwide.
- Participated in **security-focused code reviews**, addressing authentication gaps and input validation issues in management APIs serving thousands of deployed storage arrays.
- Contributed to **reusable component libraries** shared across product teams, reducing duplication and standardizing error handling and API response patterns.

- Worked cross-functionally with product and support engineering to **refine feature requirements**, estimate delivery complexity, and prioritize reliability improvements based on customer impact.
- Mentored junior engineers on **distributed system design**, test-driven development, and performance analysis practices within an onsite agile engineering organization.

Software Engineer

Mar/2009 - Jun/2013

Spatial Corp

Onsite | Broomfield, Colorado

- Built **performance-critical components** of a commercial 3D modeling SDK used by CAD and engineering applications worldwide, optimizing computational geometry algorithms for memory efficiency and execution speed.
- Designed **maintainable, testable APIs** exposing modeling kernels to Java and .NET client applications, enabling third-party developers to integrate spatial operations into enterprise workflows.
- Implemented **automated unit and integration tests** for core modeling libraries, catching regressions early and supporting continuous integration across Windows and Linux build targets.
- Collaborated with product management and customer support to **diagnose production defects** reported by OEM partners and deliver targeted fixes under tight release deadlines.
- Refactored legacy modules to improve **code quality and modularity**, reducing cyclomatic complexity and enabling parallel development across a team of eight engineers.
- Conducted **peer code reviews** with constructive feedback on API design, error handling, and backward compatibility constraints for a long-lived commercial SDK.
- Researched and prototyped **emerging rendering techniques** to improve visualization performance for large assembly models, presenting findings that influenced the next product roadmap.
- Supported **agile iteration cycles** by breaking down complex geometry features into deliverable stories with clear acceptance criteria and measurable performance benchmarks.

Application Developer

Mar/2006 - Feb/2007

Verizon Business

Remote | New York

- Developed **enterprise web applications** supporting business customer provisioning workflows, contributing Java servlets and frontend components under senior developer guidance.
- Wrote **unit tests and integration checks** for billing and account management features, helping maintain release quality in a regulated telecommunications environment.
- Assisted in **debugging production defects** by analyzing application logs and reproducing customer-reported issues in staging environments.
- Participated in **code reviews and team standups**, adopting agile practices and software development standards within a distributed engineering team.
- Implemented **RESTful service integrations** connecting internal CRM systems with customer-facing portals, ensuring data consistency across backend stores.
- Documented **API usage patterns and deployment procedures** for handoff to operations teams, improving onboarding efficiency for new team members.

EDUCATION

University of Colorado Colorado Springs

Bachelor of Science, Computer Science

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SKILLS

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Redux, state management, responsive UI design

Backend: Node.js, TypeScript, Java, Spring Boot, Kotlin, Python, RESTful API design, microservices

Databases: PostgreSQL, query optimization, schema design, SQL, distributed data operations

Cloud & DevOps: AWS (EKS, ECR, SSM, S3), Docker, Kubernetes, CI/CD, containerized deployments

Monitoring & Tools: Structured logging, metrics instrumentation, observability, Git, Jira, Claude Code, Cursor, GitHub Copilot

Workflow & PM: Agile/Scrum, cross-functional collaboration, code review, technical estimation, sprint planning